

EDUCATION

University of North Carolina at Chapel Hill, Chapel Hill, NC
Bachelor of Science in Biology (GPA: 3.94)

August 2025 - May 2029

Honors and Awards

Dean's List

Fall 2025

Buckley Public Service Scholar

Spring 2026 - Present

Honors Carolina

Fall 2025 – Present

USA Biology Olympiad – Semifinalist (Top 10% of the country)

2023-2024 & 2024-2025

RESEARCH STATEMENT

Through developing microbial co-culture organoid models of the intestinal mucosa, I aim to contribute to the scientific understanding of mucosal cells and microbial dysregulations that directly promote IBD pathogenesis. My future research interest is focused on investigating how Paneth cell dysfunction and subsequent microbial overgrowth can be regulated to improve IBD treatment from current immunosuppressive medications to therapeutics that target the cellular origin of disease.

RESEARCH EXPERIENCE

International Genetic Engineering Machine

January 2026 – Present

Wet Lab Member

- Developing probiotic therapy that targets adenoma stage in colorectal cancer, under the Koji Sode BME lab
- Produce two genetic models within the probiotic: 1) low and continuous production of IAA to support AhR activity and maintain epithelial and immune homeostasis 2) activated response to early adenoma conditions via production of indole-3-aldehyde and expression of kynureninase to promote epithelial repair and reduce immune suppression respectively
- Currently working with team to finalize project and develop experiment protocols

The Osteoarthritis Prevention Study (TOPS), UNC Thurston Arthritis Research Center.

August 2025 – Present

Undergraduate Research Intern (PI: Leigh Callahan, Lab Manager: Courtney Armitano-Lago)

- Part of a long-term cohort study to identify how diet and exercise regimens affect the progression of knee pain and osteoarthritis in female 50+ population.
- Conducted 300+ patient phone screening calls and preliminary data collection.
- Working on PatientIQ and OnCore platforms.
- Currently verifying demographic data accuracy across multiple platforms.
- Attended TARC Annual Research Conference in November 2025

TEACHING EXPERIENCE

Biology Department Peer Mentor

Peer Mentor for Biology 104 (Dr. Mara Evans)

January 2026 – Present

- Interacting with students during class to encourage further discussion, answer questions, and promote collaboration
- Conducting weekly tutoring sessions for students with open question, exam review format, and cultivation of study skills plan.

ScioVirtual Nonprofit

June 2023 – Present (Summers only)

Public Health Instructor

- Worked with teaching team to curate public health curriculum for summer classes.
- Taught classes of 50+ elementary/middle school students across the US.
- Created supplemental teaching resources, review material and interactive activities. Incorporated multiple student learning styles
- Projected to be Public Health Senior Instructor for summer of 2026

CLINICAL EXPOSURE AND SHADOWING

Shadowing of UNC Gastroenterologist Dr. Jonathan Hansen MD-PhD at Eastowne GI Clinic and Hillsborough GI Procedure Unit

February 2026 – March 2026

- 10 shadowing hours observing diagnosis and treatment process for adults with Crohn's disease and Ulcerative Colitis, and endoscopy and colonoscopy procedures.
- Observed drug choice and treatment planning based on clinical presentation of IBD and patient physical and emotional health history.

VOLUNTEER AND SERVICE WORK

Volunteer in Emergency Department, UNC Hospitals

January 2026 – Present

Undergraduate Volunteer

- Rounding on patients to check comfort levels and communicate patient concerns to nurses and physicians
- Assisting the cleanup and maintenance of procedural and supply carts

Remote Area Medical Volunteer Corps, UNC Chapter

August 2025 – Present

Community Host Group Member

- Volunteered (40+ hours) in free health clinics for underserved and uninsured populations in remote locations. Volunteered in Luray, VA (September 2025) and Henderson, NC (March 2026)
- Served in the Vision Core: conducted visual acuity tests, auto-refraction images, and lensometry measurements for 100+ patients.
- Served in Medical Core: aided in patient triage, and patient diagnosis and treatment history data input
- Aided in planning the Day of Care event at YMCA (Chapel Hill) for local homeless population. Directly collaborated with Interfaith Food Shuttle to attain 50 shelf-stable meals. Cooked and served hot meals to community members.

Student Health Action Coalition (SHAC) at Carrboro, NC

August 2025 – Present

- Served in free health clinic for underserved and uninsured population in Carrboro and surrounding areas. Completed patient intake, room cleanings, data input of diagnosis, treatment recommendations, and patient satisfaction surveys.

National Marrow Donation Program – UNC Chapter

Fundraising Committee Lead

August 2025 –

Present

- Volunteered (30+ hours) in tabling events to swab potential donors, assisting in expanding the national marrow donation registry
- Leading the fundraiser committee to plan events such as Bingo night and benefit nights with local organizations
- Taking active role in educating UNC students on the life-saving importance of marrow donation

MUSIC EXPERIENCE

Indian Classical Carnatic Music

2013 - Present

Vocalist

- Trained in Carnatic classical vocals
- Performed in group and solo concerts at venues such as Cleveland Carnatic Tyagaraja Festival and Srivari Balaji Temple
- Future plans: join and perform with the *UNC Bharat Sangeet Team*

ADDITIONAL EXPERIENCE

- Work Experience: None yet
- Internships: None yet
- Publications/presentations: None yet
- Clubs/Athletics: None yet

SKILLS

Website Design through HTML + CSS

Relevant Coursework: How Cells Function (BIOL 103), Biodiversity Honors (BIOL 104H), Cell Biology Honors (BIOL 240H), Analytical Chemistry (CHEM 241), Analytical Chemistry Lab (CHEM 241L)

Languages: Tamil (Native), Spanish (Proficient) – Seal of Biliteracy in NJ (2024)